

LXCOS: Unleashing the Power of LXC to Build PaaS Solutions



LXC or Linux Containers is virtualisation at the operating system level, which can be used to run multiple isolated containers or Linux systems on a single host. The LXCOS operating system, crafted on Chef, is versatile and can easily use pre-baked templates mainly for Web based CMS.

One are the days when we had to provision an entire operating system and install it with all the prerequisites to host an application. Thanks to virtualisation and the even more trendy containerisation, the whole provisioning, sandboxing and build release engineering has turned into an easy task.

In this modern age of Docker, CoreOS, etc, I would like to walk readers through a basic, finely crafted container platform built on pure LXC. But first, let's get familiar with some fundamental facts about LXC and the underlying concepts before getting introduced to LXCOS, its use cases and how it helps in building an internal PaaS (Platform as a Solution).

Facts about LXC

- LXC is within the kernel. You don't have to recompile a kernel. If you have Linux, you have LXC.
- LXC is operating system level virtualisation, with which you can run numerous containers on a single Linux host.
- The Linux kernel has something in its core called

cgroups, using which you can restrict resources like memory, CPU, networks, etc. On top of that, it also has *namespace isolation* which completely isolates applications within containers.

- LXC was the primary container engine for Docker before it released *libcontainer*.

In a nutshell, LXC is lightweight virtualisation with which a lot of things can be done, including a simple PaaS solution.

Consider a project in which there are several people working on different aspects. Some are writing code, some testing it and there are release managers or build engineers who send the application to production. All these people require considerable server resources to work and collaborate on. They also need their work to be isolated, in order not to disturb each other's work. So imagine a system where you can provision the LXC containers, create development environments, deploy code onto them, and sync the databases across the environments. This gives you an internal PaaS solution with much of the TechOps/ DevOps tasks automated.